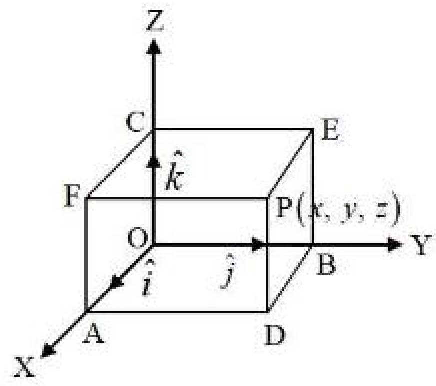
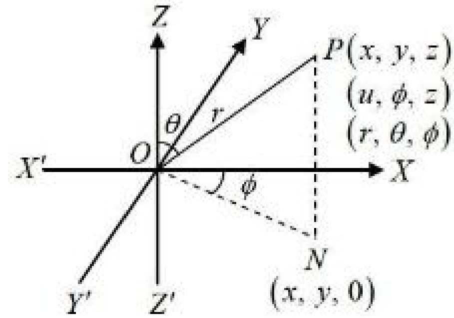
3D CO-ORDINATE GEOMETRY

- **Cartesian Co-ordinate**: Let O be a fixed point, known as origin and let OX, OY and OZ be three mutually perpendicular lines taken as x, y and z axis respectively in such a way that they form a right handed system.





The plane XOY, YOZ and ZOX are known as xy, yz and zx plane respectively. Here $OA = x,\ OB = y,\ OC = z$. The three co-ordinate planes XOY, YOZ and ZOX divides space into eight parts and these parts are called as octants. Octants are $OXYZ,\ OX'YZ,\ OXY'Z,\ OXYZ', OX'Y'Z,\ OXY'Z', OX'YZ',\ OX'Y'Z'$.

- **Cylinderical Co-ordinates**: If the rectangular cartesian co-ordinate of P are (x, y, z) then those of N are $(x, y, 0)$ and the relation is given by:
  $x = u\cos$¤, $y = u\sin$¤, $z = z$, where $u^2 = x^2 + y^2$ and ¤ $= \tan^{-1}(y/x)$.
  The cyclinderical co-ordinate of P is $(u,$ ¤$, z)$.
- **Spherical Polar co0ordinates**: The measures of quantity $r,$ ¯, ¤ are known as spherical or three dimensional polar co-ordinates of the point P. If the rectangular cartesian co-ordinate of P are (x, y, z), then the spherical co-ordinate of P is $(r,$ ¯, ¤$)$.
  $x = r\sin$¯ cos¤, $y = r\sin$¯ sin¤, $z = r\cos$¯ , where $r^2 = x^2 + y^2 + z^2$, ¯ $= \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right)$
  and ¤ $= \tan^{-1}(y/x)$.
- **Distance Formula**: The distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$.
- **Section Formula**:

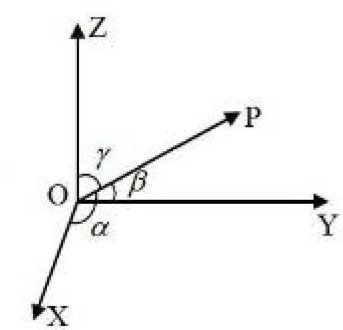
  (1) **Internal Division**: If a point (x, y, z) divides the line joining (x_1, y_1, z_1) and (x_2, y_2, z_2) internally in the ratio $m : n$, then $x = \frac{mx_2 + nx_1}{m + n},\ y = \frac{my_2 + ny_1}{m + n},\ z = \frac{mz_2 + nz_1}{m + n}$.

(2) **External Division**: If a point (x, y, z) divides the line joining (x_1, y_1, z_1) and (x_2, y_2, z_2) externally in the ratio $m:n$, then $x = \dfrac{mx_2 - nx_1}{m-n}$, $y = \dfrac{my_2 - ny_1}{m-n}$, $z = \dfrac{mz_2 - nz_1}{m-n}$.

(3) **Mid point**: If a point (x, y, z) is the midpoint of the line joining (x_1, y_1, z_1) and (x_2, y_2, z_2), then $x = \dfrac{x_2 + x_1}{2}$, $y = \dfrac{y_2 + y_1}{2}$, $z = \dfrac{z_2 + z_1}{2}$.

- **Centroid of Triangle**: If (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) are the vertices of triangle then co-ordinates of centroid is $x = \dfrac{x_1 + x_2 + x_3}{3}$, $y = \dfrac{y_1 + y_2 + y_3}{3}$, $z = \dfrac{z_1 + z_2 + z_3}{3}$.
- **Centroid of Tetrahedron**: If (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) and (x_4, y_4, z_4) are the vertices of tetrahedron then co-ordinates of centroid is $x = \dfrac{x_1 + x_2 + x_3 + x_4}{4}$, $y = \dfrac{y_1 + y_2 + y_3 + y_4}{4}$, $z = \dfrac{z_1 + z_2 + z_3 + z_4}{4}$.
- **Area of Triangle**: Let (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) are the vertices of triangle, then $\Delta_x = \left| \dfrac{1}{2} \begin{vmatrix} y_1 & z_1 & 1 \\ y_2 & z_2 & 1 \\ y_3 & z_3 & 1 \end{vmatrix} \right|$, $\Delta_y = \left| \dfrac{1}{2} \begin{vmatrix} z_1 & x_1 & 1 \\ z_2 & x_2 & 1 \\ z_3 & x_3 & 1 \end{vmatrix} \right|$, $\Delta_z = \left| \dfrac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \right|$.

  Area of triangle = $\Delta = \sqrt{\Delta^2{}_x + \Delta^2{}_y + \Delta^2{}_z}$.
- **Condition of Collinearity**: Three points (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) are collinear if $\dfrac{x_1 - x_2}{x_2 - x_3} = \dfrac{y_1 - y_2}{y_2 - y_3} = \dfrac{z_1 - z_2}{z_2 - z_3}$.
- **Volume of Tetrahedron**: If (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) and (x_4, y_4, z_4) are the vertices of tetrahedron then volume of the tetrahedron is $V = \left| \dfrac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} \right|$.
- **Direction Cosines (dcs)**:



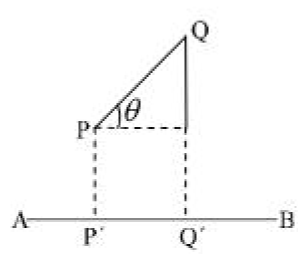
Let a vector $\overrightarrow{OP}$ (or lines OP) makes angles α, β, γ with x, y and z axis respectively. Then $\cos\alpha, \cos\beta, \cos\gamma$ are called as direction cosines of the vector $\overrightarrow{OP}$ and these are denoted by $l \cos\alpha,\ \ m = \cos\beta,\ \ n = \cos\gamma$.

(a) Direction cosines of x axis are (1, 0, 0).
(b) Direction cosines of y axis are (0, 1, 0).
(c) Direction cosines of z axis are (0, 0, 1).
(d) If $P(x, y, z)$ is a point in space, $\overrightarrow{OP} = \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then projection of $\vec{r}$ on x, y and z axis are $l|\vec{r}|,\ m|\vec{r}|,\ n|\vec{r}|$.
(e) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1 \Rightarrow l^2 + m^2 + n^2 = 1$.
(f) $\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}}$.

- **<u>Direction Ratios (drs)</u>**: A set of three numbers a, b, c (not all zero) is said to be direction ratios of a line OP or vector $\overrightarrow{OP}$ if $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$, where l, m, n are direction cosines of the line OP or vector $\overrightarrow{OP}$.

(a) If a, b, c are the direction ratios of a line, then direction cosines are given by: $l = \pm\frac{a}{\sqrt{a^2 + b^2 + c^2}},\ m = \pm\frac{b}{\sqrt{a^2 + b^2 + c^2}},\ n = \pm\frac{c}{\sqrt{a^2 + b^2 + c^2}}$.
(b) A line can have many sets of direction ratios which are proportional to each other.
(c) If $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are two points, then direction ratios of the line PQ is given by $(x_2 - x_1, y_2 - y_1, z_2 - z_1)$.
(d) If direction ratios of two lines are (a_1, b_1, c_1) and (a_2, b_2, c_2), then angle between them is given by $\theta = \cos^{-1}\left(\frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}\right)$.
Given two lines are perpendicular to each other if $a_1a_2 + b_1b_2 + c_1c_2 = 0$.
Given two lines are parallel to each other if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

- **<u>Projection</u>**: Projection of a line joining the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on another line AB whose direction cosines are l, m, n, is
Projection of PQ = $P'Q' = |(x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n|$.



Plane:

- **Equation of Planes**:

  (a) **General Equation**: If a, b, c are the direction ratios of normal to the plane, then equation of plane in Cartesian form is $ax + by + cz + d = 0$.
  Vector equation of plane is $\vec{r} \cdot \vec{n} = p$, where $\vec{n}$ is vector perpendicular to the plane.

  (b) **Normal Form of Plane**: If l, m, n are direction cosines of the normal to the plane and p be the length of perpendicular from the origin to the plane, then equation of plane in Cartesian form is $lx + my + nz = p$.
  Vector equation of plane in normal form is $\vec{r} \cdot \hat{n} = p$, where $\hat{n}$ is unit vector perpendicular to the plane.

  (c) **Intercept form of a Plane**: If a plane makes intercepts a, b, c on the axes of co-ordinates, then equation of plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

  (d) **Equation of a plane through a given point**: Equation of plane passes through a point $P(x_1, y_1, z_1)$ and a, b, c are the direction ratios of normal to the plane in Cartesian form is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.
  Vector form is $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$.

  (e) **Equation of a plane through three given points**: Equation of plane passes through three non-collinear points (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) is $\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$.

- **Equation of Systems of Planes**:

  (a) Equation of system of planes parallel to the plane $ax + by + cz + d = 0$ is $ax + by + cz + k = 0$, where k is a parameter.

  (b) Equation of system of planes perpendicular to the line $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$ is $ax + by + cz + k = 0$, where k is a parameter.

  (c) Equation of system of planes passes through the intersection of two planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is $(a_1x + b_1y + c_1z + d_1) + k(a_2x + b_2y + c_2z + d_2) = 0$

  (d) Equation of system of planes passes through the point $P(x_1, y_1, z_1)$ is $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$.

- **Angle Between two Planes**: Angle between two planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is given by $^{-} = \cos^{-1}\left(\frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}} \right)$.
  Two planes are perpendicular if $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

Two planes are parallel if $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$.

- **Two sides of a plane**:

  (a) Two points $P(x_1,y_1,z_1)$ and $Q(x_2,y_2,z_2)$ lies on the same side of the plane $ax+by+cz+d=0$ if $a_1x+b_1y+c_1z+d=0$ and $a_2x+b_2y+c_2z+d=0$ are of same sign.

  (b) Two points $P(x_1,y_1,z_1)$ and $Q(x_2,y_2,z_2)$ lies on the opposite side of the plane $ax+by+cz+d=0$ if $ax_1+by_1+cz_1+d=0$ and $ax_2+by_2+cz_2+d=0$ are of different sign.

- **Length of the perpendicular from a point to a Plane**: The perpendicular distance of the point $P(x_1,y_1,z_1)$ from the plane $lx+my+nz=p$ is $|p-lx_1-my_1-nz_1|$, where l, m, n are direction cosines of the normal to the plane and p be the length of perpendicular from the origin to the plane.
  The perpendicular distance of the point $P(x_1,y_1,z_1)$ from the plane $ax+by+cz+d=0$ is $\left|\frac{ax_1+by_1+cz_1+d}{\sqrt{a^2+b^2+c^2}}\right|$.

- **Distance between two parallel planes**: Distance between two parallel planes $ax+by+cz+d_1=0$ and $ax+by+cz+d_2=0$ is $\left|\frac{d_2-d_1}{\sqrt{a^2+b^2+c^2}}\right|$.

- **Equation of Bisector of angle between two Planes**: Let $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ are two planes, then equation of plane bisecting the angle between the given two planes which contains origin is $\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}}=\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_1^2+b_1^2+c_1^2}}$ and which do not contain origin is $\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}}=-\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_1^2+b_1^2+c_1^2}}$.

  (a) If angle between bisector plane and one of the plane is less than 45^0, then it is acute angle bisector otherwise it is obtuse angle bisector.

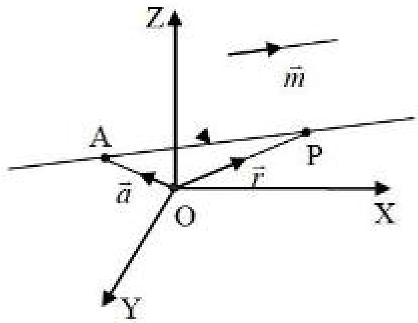
  (b) If $a_1a_2+b_1b_2+c_1c_2<0$, then origin lies in the acute angle between the given planes provided d_1, d_2 are of same sign.

  (c) If $a_1a_2+b_1b_2+c_1c_2>0$, then origin lies in the obtuse angle between the given planes.

Straight Lines:

- **Equation of Straight Lines**:

  (a) **Vector Equation**: Vector equation of a line passing through a point A whose position vector is $\vec{a}$ and is parallel to a vector $\vec{m}$ is given by $\vec{r}=\vec{a}+ª\vec{m}$, where ª is constant.



(b) **Cartesian Equation**: If a straight line passes through a given point $P(x_1, y_1, z_1)$ and has direction cosines l, m, n, then equation of the line is $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$. The coordinates of any point on it satisfy the equation $x = x_1 + lr,\ y = y_1 + mr,\ z = z_1 + nr$.

The equation of line through a given point $P(x_1, y_1, z_1)$ and has direction ratios a, b, c is $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=\lambda$ called as symmetric form of the line. The coordinates of any point on it satisfy the equation $x = x_1 + a\lambda,\ y = y_1 + b\lambda,\ z = z_1 + c\lambda$.

- **Equation of line through two given points**:

(a) **Vector Form**: Let position vector of two points are $\vec{a}$ and $\vec{b}$, then equation of the line passes though the two points is $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$.

(b) **Cartesian Form**: Equation of line passes through the two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is $\frac{x-x_1}{x_2-x_1}=\frac{y-y_1}{y_2-y_1}=\frac{z-z_1}{z_2-z_1}$. The coordinates of any point on the line PQ satisfy the equation $x=\frac{\lambda x_2+x_1}{\lambda+1},\ y=\frac{\lambda y_2+y_1}{\lambda+1},\ z=\frac{\lambda z_2+z_1}{\lambda+1}$.

- **Changing unsymmetrical form of a line into symmetrical form**: The unsymmetrical form of a line is $ax+by+cz+d=0,\ a'x+b'y+c'z+d'=0$, it can be changed to symmetrical form as: $\frac{x-\frac{bd'-b'd}{ab'-a'b}}{bc'-b'c}=\frac{y-\frac{da'-d'a}{ab'-a'b}}{ca'-c'a}=\frac{z}{ab'-a'b}$.

- **Angle between two lines**:

(a) **Vector Form**: Let two lines are $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$, then angle between them $\theta = \cos^{-1}\left(\frac{\vec{b}_1\cdot\vec{b}_2}{|\vec{b}_1|\,|\vec{b}_2|}\right)$.

(i) Two lines are parallel if $\vec{b}_1 = k\vec{b}_2$.

(ii) Two lines are perpendicular if $\vec{b}_1\cdot\vec{b}_2 = 0$.

(b) **Cartesian Form**: Let two lines are $\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2}=\frac{y-y_2}{b_2}=\frac{z-z_2}{c_2}$

then angle between them $\theta=\cos^{-1}\left(\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right)$.

(i) Two lines are parallel if $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$.

(ii) Two lines are perpendicular if $a_1a_2+b_1b_2+c_1c_2=0$.

- **Equation of line parallel to a given line**:

(a) **Vector Form**: Given equation of line is $\vec{r}=\vec{a}+\lambda\vec{b}_1$, then equation of line parallel to the line passes through the point $\vec{a}_2$ is $\vec{r}=\vec{a}_2+\mu\vec{b}$.

(b) **Cartesian Form**: Given equation of line is $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$, then equation of line parallel to the line passes through the point $Q(x_2,y_2,z_2)$ is $\frac{x-x_2}{a}=\frac{y-y_2}{b}=\frac{z-z_2}{c}$.

- **Equation of line perpendicular to two given line**:

(a) **Vector Form**: Given equation of two lines are $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$, then equation of line perpendicular to both the lines $\vec{r}=\vec{a}_3+\mu\vec{b}_3$, where $\vec{b}_1\cdot\vec{b}_3=0$ and $\vec{b}_2\cdot\vec{b}_3=0$ or $\vec{b}_3=\vec{b}_1\times\vec{b}_2$.

(b) **Cartesian Form**: Given equation of two lines are $\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2}=\frac{y-y_2}{b_2}=\frac{z-z_2}{c_2}$, then equation of line perpendicular to both the lines is $\frac{x-x_3}{a}=\frac{y-y_3}{b}=\frac{z-z_3}{c}$ where $aa_1+bb_1+cc_1=0$ and $aa_2+bb_2+cc_2=0$.

- **To find whether two given lines cut each other or not**:

(a) **Vector Form**: Let two lines are $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$. If two lines intersect, then for some value of λ,μ both the lines these points must coincides. Hence $\vec{a}_1+\lambda\vec{b}_1=\vec{a}_2+\mu\vec{b}_2$. Equate the components of $\hat{i},\hat{j},\hat{k}$. Solve any two equations for λ,μ. Put these values in the third equation. If it satisfies the third equation, then both the lines intersect, otherwise not. To find the point of intersection put the value of λ,μ in the given equation.

(b) **Cartesian Form**: Let two lines are $L_1 : \dfrac{x-x_1}{a_1} = \dfrac{y-y_1}{b_1} = \dfrac{z-z_1}{c_1}$ and $L_2 : \dfrac{x-x_2}{a_2} = \dfrac{y-y_2}{b_2} = \dfrac{z-z_2}{c_2}$. Any point of line L_1 is $(x_1 + a_1 ª,\ y_1 + b_1 ª,\ z_1 + c_1 ª)$ and on line L_2 is $(x_2 + a_2 «,\ y_2 + b_2 «,\ z_2 + c_2 «)$. If lines intersect these two points must coincide for some values of $ª$, $«$. Equate the corresponding co-ordinates. Solve any two equations for $ª$, $«$. Put these values in the third equation. If it satisfies the third equation, then both the lines intersect, otherwise not. To find the point of intersection put the value of $ª$, $«$ in the points.

- **To find foot of the perpendicular from a point on a line**:

(a) **Vector Form**:

Let a line is $\vec{r} = \vec{a} + ª\vec{b}$ and a point is $A(\vec{a}_1)$.

(i) Find Position vector of any point on the line is $P(\vec{a} + ª\vec{b})$.

(ii) Find $\overrightarrow{AP} = \vec{a} + ª\vec{b} - \vec{a}_1$

(iii) Find value of $ª$, from $\overrightarrow{AP} \cdot \vec{b} = 0$ as P is the foot of the perpendicular.

(iv) Put the value of $ª$ in $P(\vec{a} + ª\vec{b})$ to get the foot of the perpendicular.

(v) Find length of perpendicular = $\left|\overrightarrow{AP}\right|$.

(vi) Find equation of perpendicular as $\vec{r} = \vec{a} + «(\overrightarrow{AP})$.

(b) **Cartesian Form**:

Let a line is $\dfrac{x-x_1}{a} = \dfrac{y-y_1}{b} = \dfrac{z-z_1}{c}$ and a point $A(x_2, y_2, z_2)$.

(i) Let $\dfrac{x-x_1}{a} = \dfrac{y-y_1}{b} = \dfrac{z-z_1}{c} = ª$.

(ii) Any point on the line is $P(x_1 + aª,\ y_1 + bª,\ z_1 + cª)$.

(iii) Find drs of AP = $(x_1 + aª - x_2,\ y_1 + bª - z_2,\ z_1 + cª - z_2)$.

(iv) Find the value of $ª$ from $a(x_1 + aª - x_2) + b(y_1 + bª - z_2) + c(z_1 + cª - z_2) = 0$ as P is the foot of the perpendicular.

(v) Put the value of $ª$ in $P(x_1 + aª,\ y_1 + bª,\ z_1 + cª)$ to get the foot of the perpendicular.

(vi) Find length of perpendicular = $|AP|$.

(vii) Equation of perpendicular is $\dfrac{x-x_2}{x_1 + aª - x_2} = \dfrac{y-y_2}{y_1 + bª - z_2} = \dfrac{z-z_2}{z_1 + cª - z_2}$.

- **Skew Lines**: **skew lines** are two **lines** that do not intersect and are not parallel.

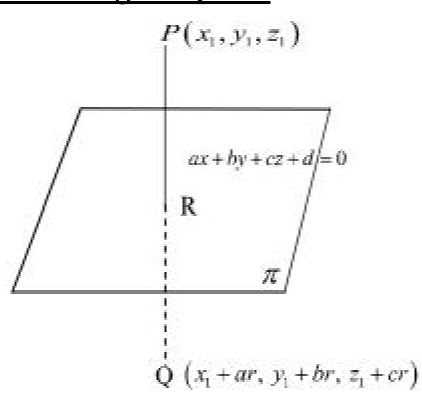
- **Lines of Shortest Distance**: If there are two skew lines, then the straight line which is perpendicular to both the lines is called as Lines of Shortest Distance. There is one and only one line perpendicular to both the lines.
- **Shortest distance between two skew lines**:

  (a) **Vector Form**: Let two skew lines are $\vec{r}=\vec{a}_1+\text{a}\vec{b}_1$ and $\vec{r}=\vec{a}_2+\text{«}\vec{b}_2$, then shortest distance between them is $\left|\dfrac{(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)}{|\vec{b}_1\times\vec{b}_2|}\right|$.

  (b) **Cartesian Form**: Let two skew lines are $\dfrac{x-x_1}{a_1}=\dfrac{y-y_1}{b_1}=\dfrac{z-z_1}{c_1}$ and $\dfrac{x-x_2}{a_2}=\dfrac{y-y_2}{b_2}=\dfrac{z-z_2}{c_2}$, then shortest distance between them is :

$$d=\frac{\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(a_1b_2-a_2b_1)^2+(b_1c_2-b_2c_1)^2+(c_1a_2-c_2a_1)^2}}.$$

- **Foot of perpendicular from a point to the given Plane**: If AP be a perpendicular from a point $A(\sim,\ ,¥)$ to a given plane $ax+by+cz+d=0$, then it is normal to the plane, so that its equation is $\dfrac{x-\sim}{a}=\dfrac{y-}{b}=\dfrac{z-¥}{c}=k$. Any point P on it is $(ak+\sim, bk+\ , ck+¥)$. It lies on the given plane, so we can find the value of k and hence the point P.
- **Image of a point in a plane**:
  Let P and Q be two points and let L be a plane such that:
  (a) Line PQ is perpendicular to the plane L.
  (b) Midpoint of PQ lie on the plane L.
  Then Point P is the image of Q and Q is the image of P.
- **To find the image of a point in a given plane**:



Let a point is $P(x_1, y_1, z_1)$ and a plane is $ax+by+cz+d=0$.

(a) Write the equation of line passes through point P and normal to the plane as $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$.

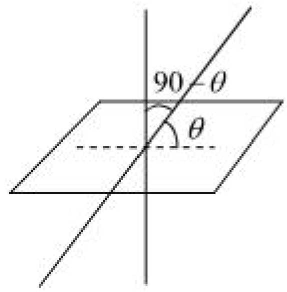
(b) Write the co-ordinate of image Q as $(x_1+a\lambda,\ y_1+b\lambda,\ z_1+c\lambda)$.

(c) Find midpoint of PQ. Midpoint of PQ lies on the plane and hence it satisfies the equation of plane.

(d) Then find the value of λ.

(e) Find the coordinate of Q by putting the value of λ in $(x_1+a\lambda,\ y_1+b\lambda,\ z_1+c\lambda)$.

- **Angle between a line and a plane**:



(a) **Vector Form**: Let a plane is $\vec{r}\cdot\vec{n}=d$ and line is $\vec{r}=\vec{a}+\lambda\vec{b}$, then angle between them is given by $\theta=\sin^{-1}\left(\frac{\vec{n}\cdot\vec{b}}{|\vec{n}|\,|\vec{b}|}\right)$.

(b) **Cartesian Form**: Let equation of plane is $a_1x+b_1y+c_1z+d_1=0$ and equation of line is $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$, then angle between them is given by $\theta=\sin^{-1}\left(\left|\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|\right)$

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